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Sexual dimorphism in sensorimotor transformation of optic flow

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eLife Assessment

Hoverflies are known for their sexually dimorphic visual systems and exquisite flight behaviors. This **valuable** study reports how two types of visual descending neurons differ between males and females in their motion- and speed-dependent responses, yet surprisingly, the behavior they control lacks any sexual dimorphism. The results **convincingly** support these findings, which will be of interest for studies of visuomotor transformations and network-level brain organization.

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Abstract

Motion vision underpins a wide range of adaptive behaviours essential for individual and species survival. Some visual behaviours are sexually dimorphic, including for example male hoverfly high-speed pursuit of conspecifics, matched by improved optics, and faster photoreceptors. Other visual behaviours are monomorphic, with for example similar foraging flight speeds in male and female hoverflies. However, whether the descending neurons responsible for sensorimotor transformation of optic flow are sexually dimorphic is unknown. To redress this, we combined morphological analysis with electrophysiology of optic flow sensitive descending neurons and compared neural responses to the wing beat amplitude in tethered hoverflies. We found that while optomotor flight behaviour is largely sexually monomorphic, the underlying neural responses are sexually dimorphic, especially at higher optic flow velocities. Additionally, neural and behavioural responses to roll stimuli had a slower onset compared to lift, revealing stimulus specific encoding. Together, our findings uncover a nuanced, sex- and stimulus- dependant sensorimotor transformation, shaped by both neural architecture and behavioural demands.

Introduction

Motion vision is a fundamental sensory modality across the animal kingdom, enabling animals to navigate, to maintain a straight trajectory, to avoid collisions, and to identify prey, predators or mates. Among the most potent cues for self-motion is optic flow, the coherent motion of the entire visual field, generated by an animal's own movement through the world. In insects, the neural mechanisms underlying optic flow processing have been studied extensively (for a review, see e.g. Ref ¹), offering a powerful model for understanding how compact nervous systems extract behaviourally relevant information from dynamic visual scenes.

To generate appropriate responses to widefield optic flow, the visual input needs to be integrated across space. In flies, this spatial pooling occurs in lobula plate tangential cells (LPTCs), each matched to a particular type of self-generated optic flow². For example, neurons of the horizontal system (HS cells) respond optimally to rotations around the yaw axis, whereas neurons in the vertical system (VS cells) are tuned to pitch and roll rotations³. While LPTCs process widefield optic flow their role in the resulting behaviours^{4,5} is indirect, via optic flow sensitive descending neurons that therefore play a key role in sensorimotor transformation^{6,7}. In *Drosophila* and

blowflies, there are three main optic flow sensitive descending neurons. DNHS1, also called NDp15⁸, receives input from HS cells^{6,7} and is physiologically similar to the optic flow sensitive descending neuron type 1 (OFS DN1) in the hoverfly *Eristalis tenax*⁹. DNHS1 has been implicated in the control of head yaw movements and abdominal ruddering and may engage with the haltere motor system for flight stabilization⁶. DNOVS1, or DNp20⁸, receiving input from the ocelli and VS cells^{6,10}, is likely involved in rapid head movements, possibly facilitating gaze stabilization during flight⁶. DNOVS2, or DNp22⁸, also receives input from the ocelli but a different subset of VS cells^{6,11} and is physiologically similar to the *Eristalis tenax* optic flow sensitive descending neuron type 2 (OFS DN2)⁹. DNOVS2 is well suited to initiate the fast body saccades that support rapid re-orientation⁶ during flight in response to dynamic visual cues.

Hoverflies are interesting in the context of motion vision, which they use both to maintain a hovering stance, and to fly at high speed¹². Indeed, hoverflies show striking sexually dimorphic flight behaviour, where males establish territories which they guard rigorously¹³ to chase away any intruding insects and/or pursue conspecific females for courtship and mating^{12,14}. Accompanying this behaviour, male *E. tenax* have larger lenses than females in a dorso-frontal bright zone¹⁵, with faster motion detection and increased signal-to-noise ratio¹⁶. In many fly species, the photoreceptors in this part of the eye are also faster in males¹⁷. In hoverflies, some LPTCs are sexually dimorphic, with males having a smaller HSN receptive field¹⁸ and velocity tuning shifted to higher velocities^{15,19}. The differences in optics, photoreceptor dynamics, and LPTC receptive field size and velocity tuning, have been interpreted as required by males in the fast flight used during sexually dimorphic territorial behaviours. Interestingly, male hoverflies are also smaller than females^{20,21}, which introduces an aerodynamic component to these behavioural and sensory specializations by reducing inertia and enabling finer control over rapid flight adjustments²². However, there is no sexual dimorphism in flight speed during non-courtship activities such as foraging between flowers²³ nor when flying within the confines of an indoor arena²⁴.

To investigate the discrepancy between sexually dimorphic visual processing and sexually monomorphic behaviours, we compared the electrophysiological response characteristics and morphology of optic flow sensitive descending neurons in male and female hoverflies and linked these findings to the wing beat amplitude (WBA) of tethered hoverflies viewing similar stimuli. The optomotor responses confirm free flight observations by showing limited sexual dimorphism at speeds up to 2 m/s for translation and 200°/s for rotation. However, while neural morphology, receptive fields and direction sensitivity of the descending neurons showed minimal sex differences, there was a significant and noticeable difference in the velocity response functions between males and females, especially at higher speeds. Critically, neural differences were not only velocity dependent but also varied between stimuli (occurring only for sideslip, lift and thrust, but not roll) and neuron type. Furthermore, we found a striking difference in response onset between lift and roll in OFS DN2 and WBA, but not OFS DN1. These neuron-, stimulus-, and sex-specific differences uncovers a previously unrecognized complexity in the neural encoding of visual motion, revealing for the first time, a sex-dependent transformation from sensory input to motor output.

Results

Two distinct types of optic flow sensitive descending neurons can be identified by their receptive field location and preferred direction

Optic flow sensitive descending neurons can be readily identified by mapping their receptive field using small sinusoidal gratings^{9,15}. Based on the receptive fields of 100 reference neurons recorded from 90 male hoverflies, we found that two key parameters, the position of the receptive field centre and its preferred direction of motion, are sufficient to reliably ascertain neuron type (Fig. 1 [↗](#)). OFS DN1 has a preferred direction up and away from the midline, either leftward

(range from 137° to 171°) or rightward (range from 16° to 40°) for neurons on the left- and right-hand side of the visual field, respectively (Fig. 1a, b and c; green, Fig. 1g, h and i). OFS DN2 neurons respond preferentially to downward motion (range from 228° to 293°; Fig. 1d, e and f; yellow and orange, Fig. 1g, h and i) with the position of the receptive field centre relative to the midline separating left hand side (LHS) from right hand side neurons (RHS, Fig. 1g, h and i).

We used the maximum local motion sensitivity (LMS, Fig. 1c, f), the extent of the receptive fields (number of positions with LMS over 50%, Fig. 1c, d) and the local preferred direction (LPD, Fig. 1c, d) variance from these 100 reference neurons (grey data, Supplementary Fig. 1a, b and c) to set strict exclusion criteria (dashed red, Supplementary Fig. 1a, b and c) of the neurons used in the rest of the paper. This resulted in the exclusion of two neurons from males and seven from females (grey, Supplementary Fig. 1d, e) due to either low LMS (less than 20 spikes/s, Supplementary Fig. 1a), a small number of locations where LMS exceeded 50% of the maximum (4 positions or less, Supplementary Fig. 1b) or high LPD variance (above 30°, Supplementary Fig. 1c). A comparison of the remaining 33 male and 29 female neurons showed no significant sexual dimorphism in receptive field size for either neuron type (Supplementary Fig. 1f, unpaired t-test, $p = 0.52$ and 0.09 for width and height of OFS DN1, and $p = 0.19$ and 0.13 for width and height of OFS DN2).

Directional tuning of optic flow sensitive descending neurons exhibits limited sexual dimorphism

Some LPTCs, which are upstream of the descending neurons, show distinct sexual dimorphism, whilst others do not^{15,18}. To determine the extent of sexual dimorphism in the descending neurons, we compared the direction tuning of males and females in response to full screen stimuli. When presented with full-screen sinusoidal gratings (wavelength 7°, 5 Hz), the preferred direction of both OFS DN1 and OFS DN2 matched their receptive field preferred directions (Supplementary Fig. 1d, e and Supplementary Fig. 2c, d), i.e. up and away from the visual midline for OFS DN1 (range from 359° to 52°) or downwards for OFS DN2 (range from 273° to 297°, Fig. 2a). Whilst OFS DN1 showed no difference in preferred direction between the sexes (Watson-Williams two-sample test, $p = 0.38$), male OFS DN2 had a slightly more lateral preferred direction compared to females (Fig. 2a; median = 285.4° compared to 281.5°, Watson-Williams two-sample test, $p = 0.046$).

We next looked at the responses to 3-dimensional starfield stimuli simulating the type of optic flow that would be generated by self-motion through space²⁵. Neither the spontaneous rate nor the responses of OFS DN1 to a stationary starfield pattern differed between the sexes (circles, Supplementary Fig. 3a, two-way ANOVA, $p = 0.29$). Conversely, OFS DN2 exhibited significant sexual dimorphism, with a higher spontaneous rate and response to stationary stimuli in females compared to males (circles, Supplementary Fig. 3b, two-way ANOVA, $p < 0.001$). To determine if there was additional sexually dimorphic dependence on the type of optic flow, we subtracted the response to the stationary stimulus from the responses to each direction of the optic flow (0.5 m/s for translations and 50°/s for rotations) within each trial. As predicted from the receptive fields (Fig. 1) OFS DN1 was excited by stimuli moving either up or away from the midline, such as rightward sideslip and yaw, and upwards lift and pitch (Fig. 2b), whilst OFS DN2 showed the strongest responses to downwards lift and pitch (Fig. 2c). Both neuron types respond strongly to clockwise roll, as predicted by their receptive fields (Fig. 1a, b) and shown previously for males⁹. However, neither neuron type showed any sexual dimorphism (Fig. 2b, c, two-way ANOVA, $p = 0.81$ and 0.92 , OFS DN1 and OFS DN2, respectively).

Sexual dimorphism in optic flow descending neurons is velocity dependent

As the velocity tuning of some LPTCs is sexually dimorphic^{15,19}, we tested the responses of optic flow sensitive descending neurons using a continuous velocity step stimulus (Fig. 3, Supplementary movie 1). Six velocities for each direction of the stimuli (e.g. anticlockwise roll -10

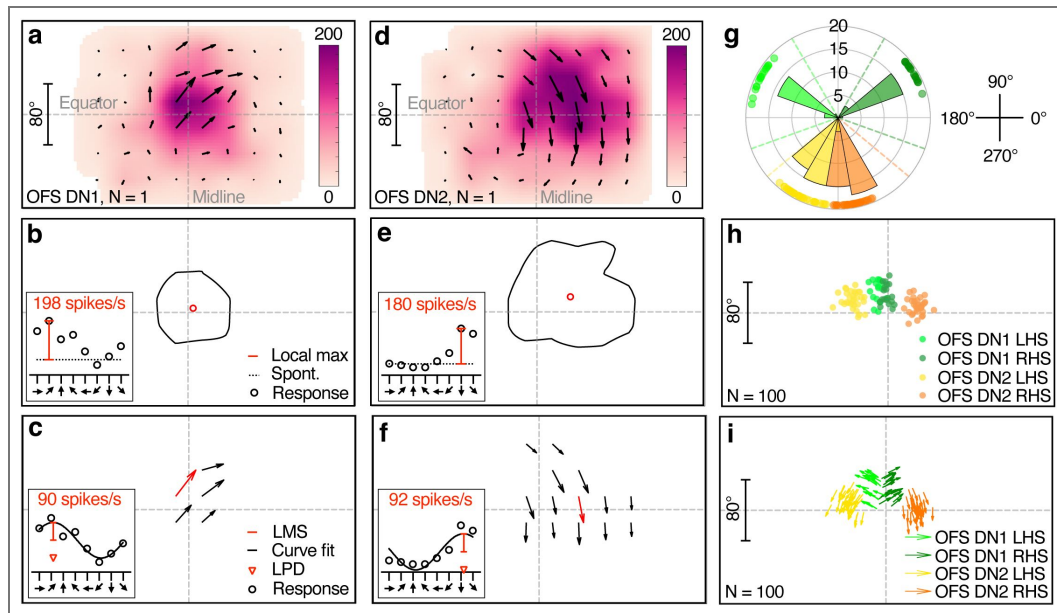


Fig. 1. Classification of optic flow sensitive (OFS) descending neurons (DN) in *Eristalis tenax*.

a Receptive field of a representative OFS DN1 recorded from a male hoverfly. Colour coding indicates the local maximum spike frequency, and the direction of the arrows shows the local preferred direction (LPD) and their length the local motion sensitivity (LMS). **b** Contour line representing the 50% receptive field boundary based on local maximum spike frequency from panel a. Inset: example response at one central location to eight directions of motion (black circles), showing local maximum spike frequency (local max, red line) above spontaneous rate (spont, black dotted line). **c** Preferred direction map of the same example neuron at locations where local motion sensitivity (LMS) exceeds 50% of the maximum (red arrow). Inset: example response at one location to eight directions of motion with a sinusoidal fit (black line), illustrating LMS (red line) and LPD (red open arrowhead). **d** Receptive field of a representative male OFS DN2. **e** 50% receptive field contour and receptive field centre of the neuron shown in panel d. **f** Preferred direction map of the same OFS DN2. **g** Distribution of receptive field preferred directions across 100 male reference neurons, colour-coded by neuron classification. Dashed lines in corresponding colours indicate the thresholds used for neuron type classification. **h** Receptive field centres of the same 100 reference neurons, using the same colour coding. **i** Relationship between preferred direction and receptive field centre for the 100 neurons.

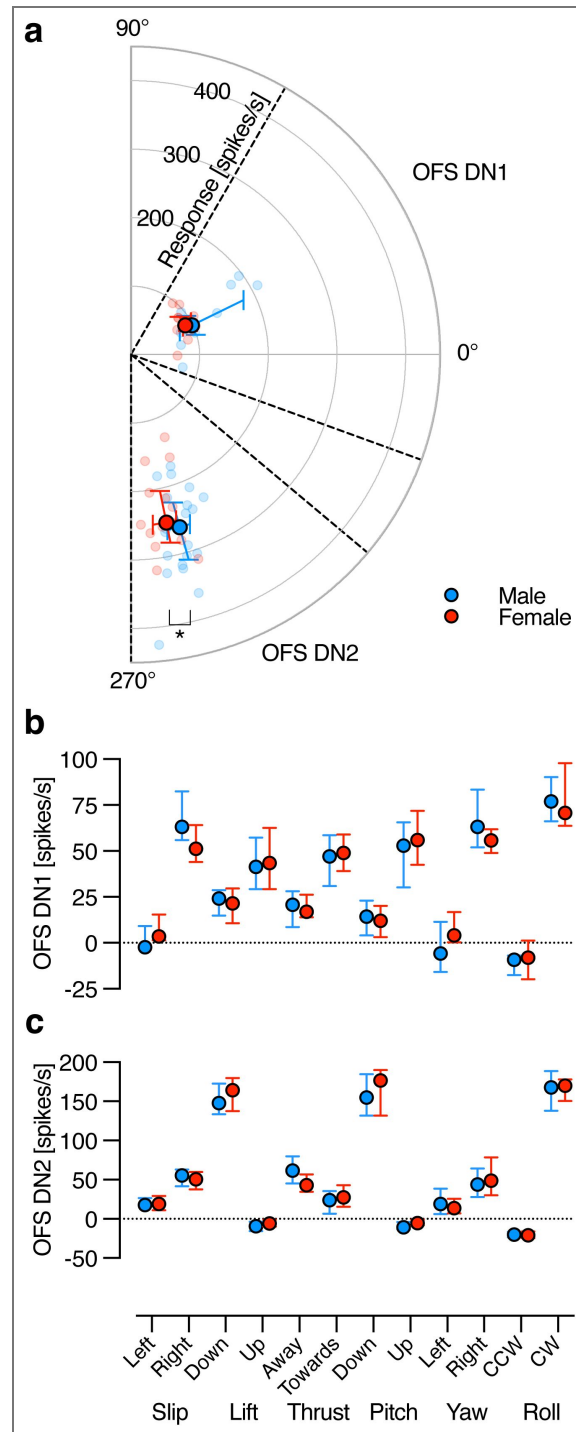


Fig. 2. Sex-based comparison of direction sensitivity in OFS DNs.

a Polar plot showing the preferred direction of male (blue) and female (red) OFS DNs in response to a full-screen, full-contrast sinusoidal grating (spatial wavelength 7°, temporal frequency 5 Hz). Individual data points represent the response amplitude and preferred direction of each OFS DN. Larger, salient circles indicate the population median, with error bars showing the interquartile range. The dashed lines indicate the directional thresholds used to classify neuron type, as OFS DN1 (male: N = 9; female: N = 12) or OFS DN2 (male: N = 20; female: N = 14). Asterisk indicates a statistically significant difference ($p < 0.05$), Watson-Williams two-sample test. **b** Comparison of male and female OFS DN1 responses to translational optic flow at 0.5 m/s: sideslip, lift and thrust; and rotational optic flow at 50 °/s: pitch, yaw and roll (male: N = 9; female: N = 12). **c** Comparison of male and female OFS DN2 responses to the same optic flow stimuli (male: N = 20; female: N = 14). Data presented as median and interquartile range.

to $-200^\circ/\text{s}$ and clockwise roll $+10$ to $+200^\circ/\text{s}$) and a stationary control were presented 3 times each in random order for 2 s each (see example, Fig. 3a).

We confirmed that the spontaneous rate and response to stationary stimuli was significantly higher in females compared to males for OFS DN2 but not for OFS DN1 (triangles, Supplementary Fig. 3b, two-way ANOVA, $p = 0.18$ and $p < 0.001$, OFS DN1 and OFS DN2, respectively). As above, we subtracted each trial's average response to the stationary stimulus from the responses to moving optic flow. The resulting data show that both neuron types exhibit sexual dimorphism to certain types of optic flow (Fig. 3e, f and Table 1). For example, the OFS DN1 response to thrust show a significant interaction between sex and velocity, with female neurons responding stronger to positive thrust compared to males (Fig. 3e and Table 1). The OFS DN2 response to sideslip was significantly different between males and females, and there was a significant interaction between sex and velocity in the responses to sideslip, lift, and thrust, with males responding stronger to sideslip, lift and thrust (Fig. 3f and Table 1).

Morphological reconstruction suggests optic flow sensitive neurons could control wing movements

By recording from the descending neurons intracellularly we could iontophoretically fill them with 3% neurobiotin following identification based on receptive field (as in Fig. 1). We found that OFS DN1 and OFS DN2 of either sex receive their input in the part of the brain where LPTCs have their output (Fig. 4). Both neuron types also have wide branches projecting to the area of the thoracic ganglion where the prothoracic and pterothoracic nerves likely get their inputs (T1 LN and PtN, Fig. 4a), suggesting they could contribute to controlling the wings and/or the forelegs. Whilst there was no indication of extensive sexual dimorphism in the structural morphology of these neurons, OFS DN2 appears to be slightly wider along the length of the cervical connective in females (Supplementary Fig. 4b). In part, this may be due to female hoverflies being bigger^{20,21} and therefore having a wider cervical connective (Supplementary Fig. 4c).

Wing beat amplitude changes are velocity dependent but not sexually dimorphic

Given that the morphological data suggest that both neuron types could control the wings (Fig. 4), we conducted behavioural experiments in an open-loop tethered flight arena using the same continuous velocity stimulus as in electrophysiology (see Fig. 3a). We tracked the wing beat amplitude (WBA) for each wing (Fig. 5a, b, Supplementary movie 2) using DeepLabCut models^{26,27} while the tethered animal was viewing different starfield velocities (Fig. 5c-f). Changes in the sum of the left and right WBA (WBAS) reflect variations in flight force generation, potentially contributing to either lift or thrust, while the difference between left and right WBA (WBAD) indicates yaw turning behaviour^{28–30} or adjustments to body orientation³¹.

The male WBAS is higher when viewing stationary stimuli compared with pre-stimulation (“pre stim”, Fig. 5c, d; two-way ANOVA followed by Uncorrected Fisher's LSD test, $p < 0.0001$, blue squares, Supplementary Fig. 3c). The male WBAS was also larger than in female hoverflies viewing stationary stimuli (two-way ANOVA followed by Uncorrected Fisher's LSD test, $p = 0.02$, Supplementary Fig. 3c). In contrast, when comparing responses to each type of optic flow, after subtracting the response to stationary stimuli, both the WBAD and WBAS depended strongly on velocity, however, evidence for sexual dimorphism was limited (Fig. 5g, h and Table 1). Indeed, only the WBAS in response to lift showed a significant sex-velocity interaction (Fig. 5g, h and Table 1).

The limited extent of sexual dimorphism observed in behaviour (Fig. 5g, h and Table 1) contrasts with the neural responses (Fig. 3e, f and Table 1). Additionally, it is difficult to align the behavioural responses to different types of optic flow with the neuronal input. For example, in response to higher clockwise roll velocities of ($100 - 200^\circ/\text{s}$), hoverflies exhibit a decreased WBAD (positive roll, Fig. 5g) and a slightly increased WBAS (positive roll, Fig. 5h) indicating either a leftward turn or an anticlockwise body correction. Similarly, both RHS neuron types give strong

Fig. 3. Velocity response functions in male and female OFS DNs.

a Example stimulus profile over time, with roll velocity on the y-axis. **b** Representative spike histogram from a single trial from a male OFS DN2, smoothed using a 100 ms square-wave filter with 0.025 ms resolution, and time-aligned to the stimulus shown in panel a. Grey shading in panels a and b highlight the analysis windows used to calculate response. **c** Example extracellular raw data traces extracted from the analysis windows in panel b, illustrating neuronal responses to roll velocities of -150, 10, and 200 °/s. **d** Average spike frequency (grey circles) was calculated for each repetition from each neuron (N = 1 neuron, n = 18 repetitions), and the median of these (black) was used for further analysis. **e** Velocity response functions of OFS DN1 in male (blue) and female (red) hoverflies in response to roll (N = 4 males, 5 females), sideslip (N = 5, 6), lift (N = 4, 4), and thrust (N = 3, 5). **f** Velocity response functions of OFS DN2 to roll (N = 10 males, 8 females), sideslip (N = 6, 7), lift (N = 6, 7), and thrust (N = 6, 7). Data in panels e and f are presented as median and interquartile range. Asterisks indicate statistically significant differences, two-way ANOVA with Šídák's multiple comparisons test (** $p < 0.01$ and **** $p < 0.0001$), see also Table 1.

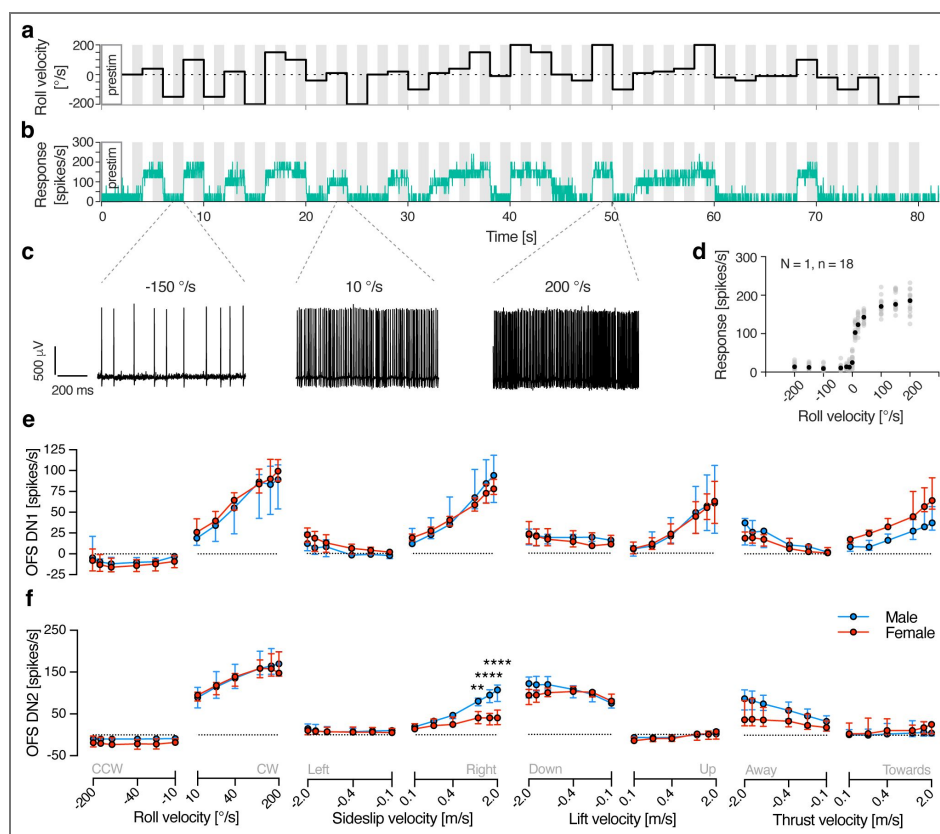


Table 1. Statistical summary of neural and behavioural velocity response functions.

Results from two-way ANOVA analyses evaluating the effects of stimulus velocity and sex on both neural (OFS DN1 and DN2) and behavioural (WBAS and WBAD) responses. Asterisks denote levels of statistical significance: *** $p < 0.001$, **** $p < 0.0001$.

Stimulus	Velocity	Sex	Interaction	Velocity	Sex	Interaction
Roll	****	0.35	0.14	****	0.31	>0.99
Sideslip	****	0.90	0.12	****	0.01	****
Lift	0.002	0.86	>0.99	****	0.18	***
Thrust	****	0.11	****	****	0.10	0.02
Roll	****	0.11	0.66	****	0.13	0.76
Sideslip	****	0.76	0.35	****	0.61	0.79
Lift	0.03	0.79	0.52	****	0.09	0.02
Thrust	0.29	0.66	0.72	***	0.57	0.83

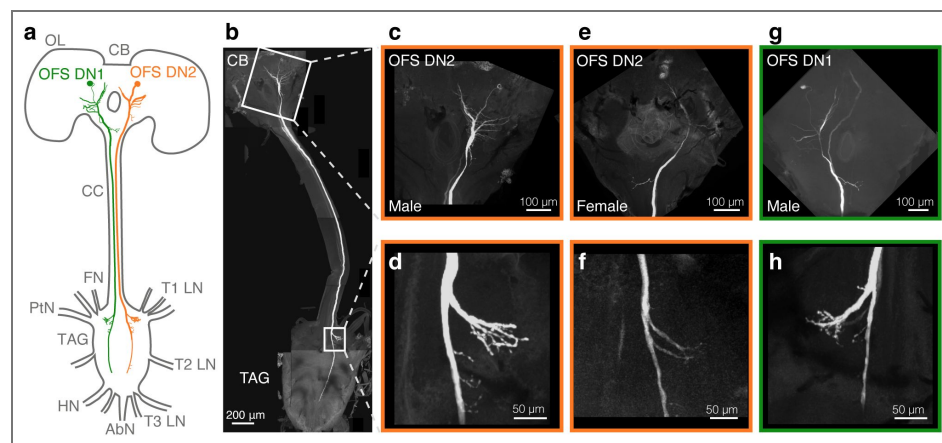


Fig. 4. Morphological reconstruction of OFS DNs.

a Schematic diagram of the hoverfly central brain and thoracic ganglia showing the projections of OFS DN1 (green) and OFS DN2 (orange). Key anatomical landmarks are labelled: OL, optic lobe; CB, central brain; CC, cervical connective; T1 LN, prothoracic leg nerve; T2 LN, mesothoracic leg nerve; T3 LN, metathoracic leg nerve; TAG, thoracic-abdominal ganglion; FN, frontal nerve; PtN, pterothoracic nerve; HN, haltere nerve; AbN, abdominal nerve. **b** Confocal image of a reconstructed male OFS DN2. White boxes indicate regions magnified in panels c and d. **c** Input dendrites of OFS DN2 around the sub-oesophageal ganglion. **d** Output projections of the same OFS DN2 neuron within the thoracic ganglia. **e** Input dendrites of a female OFS DN2. **f** Output projections of the same neuron. **g** Input dendrites of a male OFS DN1. **h** Output projections from the same neuron.

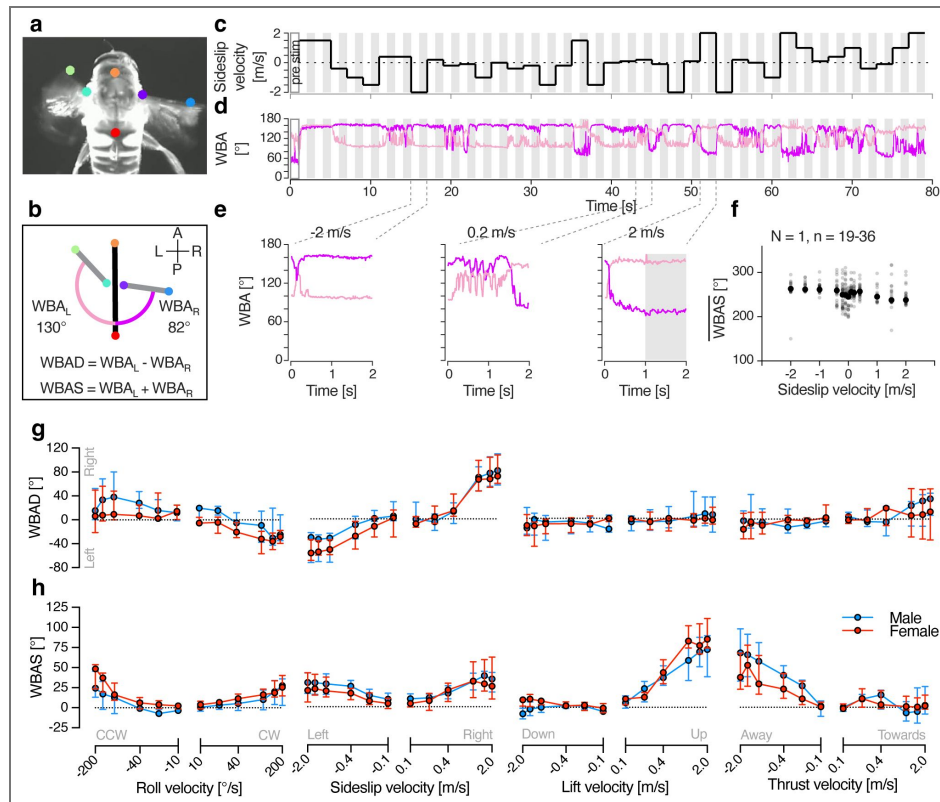


Fig. 5. WBA velocity response functions in male and female hoverflies.

a Screenshot from a video labelled with trained DeepLabCut models^{26,27,44}. Coloured dots correspond to those in panel b, used to extract the wing beat amplitude (WBA). **b** Pictogram illustrating a higher wing beat amplitude on the left wing (WBA_L) compared to the right wing (WBA_R), suggesting a turn to the right. Equations used to calculate wing beat amplitude difference (WBAD) and wing beat amplitude sum (WBAS). **c** Example stimulus with sideslip velocity on the y-axis. **d** Representative WBA for the left (pale colour) and right (salient colour) wings from a single trial, time-aligned with the stimulus shown in panel c. Grey shading in panels c and d indicate the analysis windows used to calculate WBAD and WBAS. **e** Magnified view of example WBA, showing behavioural responses to sideslip velocities of -2, 0.2, and 2 m/s. **f** Average WBAS (grey circles) calculated across repetitions ($N = 1$ animal, $n = 19-36$ repetitions). Black circles represent the median WBAS per stimulus condition, used for further analysis. **g** WBAD in male (blue) and female (red) hoverflies in response to different optic flow velocities: roll ($N = 5$ males, 7 females), sideslip ($N = 6, 6$), lift ($N = 7, 5$), and thrust ($N = 6, 5$). **h** WBAS to the same stimuli in the same animals. Data in panels g and h are presented as median and interquartile range.

responses to clockwise roll (positive roll, Fig. 3e, f). However, sideslip moving leftwards also elicits a decreased WBAD (negative sideslip, Fig. 5g) and an increased WBAS (negative sideslip, Fig. 5h) but neither neuron type responds strongly to this (negative sideslip, Fig. 3e, f).

We found that neither lift nor thrust evoke large turning responses (Fig. 5g), however, large and significant increases in the WBAS were observed in hoverflies experiencing the sensation of falling (positive lift, Fig. 5h and Table 1) or being pushed backwards (negative thrust, Fig. 5h and Table 1). Interestingly, each neuron type responds to lift and thrust in a different manner, with OFS DN1 being excited by upwards lift and approaching thrust (Fig. 3e), whereas OFS DN2 is excited by downwards lift and receding thrust (Fig. 3f).

In electrophysiology we used a horizontally oriented visual monitor (Fig. 1), whereas in behaviour the monitor was rotated 90°. To ensure that this did not substantially affect the shape of the velocity-response functions, we reduced the stimuli to the central square in both behaviour and electrophysiology. This reduction caused a slight but significant change in neuronal (central square; Supplementary Fig. 5a, two-way ANOVA, $p = 0.04$) and behavioural responses (central square; Supplementary Fig. 5b, two-way ANOVA, $p = 0.04$). However, it did not alter the shape of the velocity response function nor produce a significant interaction between velocity and screen size (two-way ANOVA, $p = 0.36$ and 0.49 , OFS DN2 and WBAS, respectively).

Response onset varies depending on stimulus type

It has been previously shown that while neuronal responses are fast²⁵, behavioural responses occur on a much slower time scale and differ depending on stimulus type^{32,33}. We here found that there is no significant effect of sex on either neuronal or behavioural response onset (Fig. 6, two-way ANOVA, $p = 0.21$, 0.79 and 0.31 , OFS DN1, OFS DN2 and WBAS, respectively). However, the onset for roll compared to lift is significantly longer for OFS DN2 and WBAS but not for OFS DN1 (Fig. 6, two-way ANOVA, $p < 0.0001$ for both OFS DN2 and WBAS compared to $p = 0.08$ for OFS DN1).

Discussion

In the hoverfly, the optics, photoreceptors, and HS cells exhibit clear sexual dimorphism^{15,16,18,19}. In line with this, our findings reveal that optic flow sensitive descending neurons also show sexually dimorphic velocity response functions (Fig. 3). Yet, despite the multi-level anatomical and neural sexual dimorphism, the wing beat amplitude remains mostly monomorphic (Fig. 5). Moreover, differences between neural and wing beat amplitude velocity response functions (Fig. 3, 5) and response latency (Fig. 6) highlight the likely involvement of downstream processing mechanisms that could reconcile sex-specific sensory encoding with conserved flight control.

Although male hoverflies pursue and capture female conspecifics at high speeds¹⁴, previously reported cruising speeds in both field²³ and indoor settings²⁴ do not differ between sexes. In addition, the median reported speeds are around 0.3 m/s^{23,24}, well below the 10 m/s reported during outdoor pursuit¹⁴. In comparison, our results show limited sex-related differences in WBA in response to optic flow speeds up to 2 m/s (Fig. 5), yet we detected pronounced sexual dimorphism in neural responses at velocities as low as 0.5 m/s (Fig. 3, Table 1). This suggests that sensory processing differences emerge at lower velocities than the velocities where motor output diverges. Moreover, the WBA response latency to roll is much longer than to lift, and much longer than the corresponding neural activity (Fig. 6), implying that temporal integration or additional circuit-level modulation may delay and refine motor execution. Together, these findings suggest a complex transformation between sex-specific sensory encoding and conserved motor behaviour, potentially mediated by downstream integration or additional alternate neural pathways that bypass the descending neurons studied here.

Both neurons project broadly within the thoracic ganglion, including in zones associated with prothoracic and pterothoracic nerve input (Fig. 4), suggesting involvement in wing and foreleg motor control. Nevertheless, there is currently no direct evidence implicating them in the control

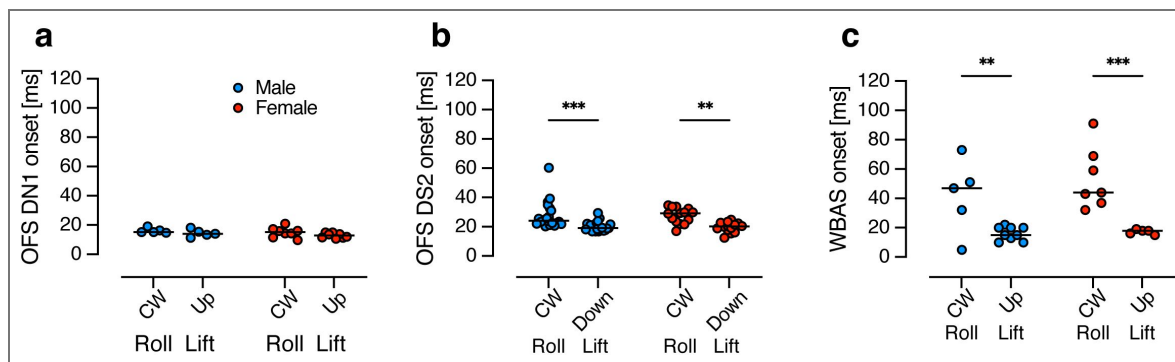


Fig. 6. Response onset to roll and lift stimuli in male and female hoverflies.

a Time to response onset in OFS DN1 to roll (+50 °/s) or lift (+0.5 m/s) stimuli in males (blue, N = 5) and females (red, N = 8). **b** Response onset in OFS DN2 to roll (+50 °/s) or lift (−0.5 m/s) measured in males (N = 20) and females (N = 14). **c** Onset of WBAS responses to roll (−200 °/s) and lift (+2 m/s) in males (N = 5 for roll, 9 for lift) and females (N = 7 for roll, 5 for lift). Data are presented as individual repeats with lines indicating median. Asterisks indicate statistically significant differences, two-way ANOVA with uncorrected Fisher's LSD test (** p < 0.01 and *** p < 0.001).

of wingbeat amplitude. Interestingly, DNOVS2 in *Drosophila*, a physiologically similar neuron to OFS DN2⁹, has been indirectly linked to rapid turning behaviour⁶. Furthermore, silencing HS and VS cells, the presumed presynaptic LPTCs to these descending neurons, results in reductions in WBA only at higher stimulus speeds³⁴. These findings suggest that HS and VS cells, along with their downstream targets, may be specialized for driving fast optomotor responses under high-speed visual motion (>180°/s), rather than broadly regulating WBA across all velocities, including those examined in this study.

Beyond their potential involvement in wingbeat amplitude modulation, OFS DNs may play a role in coordinating head and body positioning during flight, especially in contexts requiring precise visual alignment and rapid manoeuvring. Indeed, the *Drosophila* physiological homologs of OFS DN1 and OFS DN2⁹, DNHS1 and DNOVS2, have been implicated in controlling head movements, abdominal ruddering, and engagement with the haltere motor system for flight stabilization⁶. Moreover, the sexual dimorphism in neural responses (Fig. 3) could reflect an evolutionary tuning of visuomotor pathways in males, optimized for fast, directional adjustments rather than gross changes in WBA. For instance, whilst changes in WBAS are similar when generating either lift or thrust (Fig. 5), body pitch dynamically adjusts their ratio, driving behavioural variation³⁵. Thus, OFS DNs may serve as integral components of a broader flight control architecture, interfacing optic flow detection with dynamic body and head positioning systems to support complex, sex-specific behavioural outcomes, particularly in males engaging in high-speed pursuits.

Flight speed in insects results from a complex integration of multiple kinematic parameters, not solely from changes in wingbeat amplitude. Many species refine their aerodynamic output by modulating wingbeat frequency, angle of attack, wing tip trajectory, deviations from the mean stroke plane and through precise adjustments to the timing and duration of the up- and downstrokes³⁶. These control strategies support agile manoeuvring, particularly during visually guided behaviours such as the high-speed pursuits undertaken by male hoverflies. Furthermore, the smaller body size of male hoverflies compared to females^{20,21} may confer biomechanical advantages, including reduced mass, facilitating faster acceleration, heightened responsiveness, and lower metabolic costs for executing flight manoeuvres^{22,37}, which do not appear in a tethered flight set-up. Such enhanced agility is likely crucial for rapid direction changes required during courtship and may enable male hoverflies to outperform female flies without relying on increased wingbeat amplitude.

Taken together, our findings reveal significant differences between sexually dimorphic sensory encoding and conserved motor output in hoverfly flight at cruising speeds. Although OFS DNs and their upstream visual circuits display clear sex-specific tuning, wingbeat amplitude changes in response to optic flow stimuli remain relatively similar between the sexes, suggesting additional mechanisms downstream of sensory input. This likely reflects a complex interplay of biomechanical properties, multisensory integration and circuit-level modulation, each shaping and refining behavioural outcomes to meet distinct demands, preserving the consistency of low-speed manoeuvres whilst enabling sex-specific tuning during high-speed pursuits.

Methods

Animals

For all experiments, male and female *Eristalis tenax* were reared and housed as described previously²¹. Briefly, eggs were collected from females captured under permit in Wittunga Botanic Garden, Adelaide, South Australia. Upon hatching, larvae were reared in a rabbit dung slurry until third instar larvae emerged to pupate. Eclosion occurred 1-2 weeks post-pupation. Adult hoverflies were used for behavioural experiments at 17-87 days post-eclosion, for intracellular electrophysiology and subsequent morphological reconstruction at 38-54 days, and for extracellular electrophysiology at 8-204 days.

Electrophysiology

Before recording the animal was immobilised, mounted dorsal side down and secured using a mixture of beeswax and resin. A small region of cuticle was removed at the anterior end of the thorax to expose the cervical connective. If required, any excessive gut or tracheal tissue obstructing the recording site was removed and a small volume of PBS was added to prevent drying within the ventral cavity. A fine wire hook was positioned under the cervical connective for mechanical support, and a silver wire was inserted into the cavity to serve as a reference electrode and grounding wire⁹.

For extracellular recordings, a sharp tungsten microelectrode (2 MΩ, polyimide-insulated; Microprobes) was inserted into the cervical connective⁹. Signals were amplified 1000 times and band-pass filtered between 10–3000 Hz using a DAM50 differential amplifier (World Precision Instruments), followed by noise reduction with a HumBug (Quest Scientific). Data were digitized via a PowerLab 4/30 interface (ADInstruments) and acquired at 40 kHz. Spike sorting was performed in LabChart 7 Pro (ADInstruments) based on the amplitude and width of individual action potentials.

For intracellular recordings, aluminosilicate electrodes were pulled using a Sutter P-1000 micropipette puller, achieving a resistance of approximately 40–70 MΩ. Electrode tips were filled with 3% neurobiotin (Vector Laboratories), then backfilled with 1 M KCl using a syringe, leaving a small air bubble between the two solutions. Electrodes were inserted into the cervical connective for recording, and the resulting signal was amplified using an Axoclamp-2B amplifier (Axon Instruments), followed by 50 Hz noise reduction with a HumBug (Quest Scientific). Data acquisition and digitization were performed at 10 kHz using an NI USB-6210 16-bit data acquisition card (National Instruments) and the MATLAB Data Acquisition Toolbox (Mathworks), using in house software (<https://github.com/HoverflyLab/SampSamp> ).

Morphological reconstructions

Following intracellular recordings, neurons were stained iontophoretically with neurobiotin using currents in the 1 nA range for 3–12 minutes. The nervous system was then carefully dissected and fixed in 4% paraformaldehyde overnight. Tissue was incubated with a Cy3-streptavidin conjugate (1:100; Jackson ImmunoResearch) for 2 hours, then dehydrated through an ethanol series (50–100%) for 15–20 minutes per step. After washing in PBT, the tissue was cleared in RapiClear (SUNJIN Lab) and mounted with spacers. Imaging was performed using a Zeiss LSM 880 Fast Airyscan confocal microscope at the institutional microscopy facility. Neuron morphology and cervical connective width were quantified using ImageJ³⁸.

Tethered flight

Prior to flight recordings, hoverflies were tethered at a 32° angle using a beeswax–resin mixture to a small pin inserted into a hypodermic needle (BD Microlance, 23G × 1¼"). Flight was initiated by manually providing airflow for 1–10 minutes until consistent flight behaviour was observed. Once positioned at the centre of the flight arena, hoverflies were filmed from above at 100 Hz using a Sony PlayStation 3 Move Eye Camera (SLEH-00448, Sony) with the IR filter removed, and equipped with an infrared pass filter (R72 INFRARED, 49 mm, HOYA; for details see Ref²⁷). Illumination was provided by infrared LEDs inserted into USB lights (JANSJÖ LED USB lamp, IKEA) and a Musou Black (Shin Kokushoku Musou black, KOYO Orient Japan) surface was placed beneath the hoverfly to enhance contrast and minimize optical interference.

We used DeepLabCut (DLC) version 2.3.3²⁶ to train a model to track the thorax, and the peak downstroke angle, referred to as the wing beat amplitude (WBA), of the left and right wing (WBA_L and WBA_R; Fig. 5a, b ) as described previously²⁷. Briefly, we manually labelled six anatomical landmarks: the tegula and wing tip of both left and right wing, the anterior thorax, and the anterior abdomen (Fig. 5a, b ) across 16 extracted video frames per individual from four

hoverflies (2 males, 2 females). In addition to examples where the hoverfly was not flying, these frames included responses to yaw rotation and forward translation. The DLC model was trained for 300,000 iterations, yielding train and test errors of 1.2 and 1.16 pixels, respectively.

To identify potential tracking errors from DLC, we smoothed the WBA time series using MATLAB's *smooth* function with both *loess* and *rloess* methods. Because *rloess* is more resistant to outliers, we used it to detect abnormal data points. For each wing, if the absolute difference between the *loess*- and *rloess*-smoothed signals exceeded 5% for more than 1 s, the data were excluded due to suspected tracking artifacts. The smoothing was used only for error detection; all subsequent analyses were performed on the unsmoothed data. In addition, we excluded data if the WBA of either wing dropped below 40° for at least 0.5 s, as this indicated a cessation of flight. Finally, entire trials were excluded if the hoverfly was not flying for more than 50% of the trial duration.

Visual stimuli

Visual stimuli were generated using custom software (<https://github.com/HoverflyLab/FlyFly/releases/tag/v4.2.3>) written in MATLAB, incorporating the Psychophysics Toolbox^{39,40}. All screens had a refresh rate of 165 Hz and a linearized contrast with a mean illuminance of 200 Lux. For intracellular recordings, the hoverfly was placed 13 cm away from a ViewSonic screen with a resolution of 2560 × 1440 pixels, corresponding to 143° × 107° of the visual field. For extracellular recordings, the hoverfly was placed 6.5 cm away from a 2560 × 1440 pixel Asus screen, yielding a projected visual field of 155° × 138°. For behavioural recordings, the hoverfly was placed 10 cm from a vertically orientated Asus screen (1440 × 2560 pixel) producing a projected size of 118° × 142°. To evaluate the impact of different screen orientations in electrophysiology and behaviour, visual stimuli were presented either full-screen or using the central 1440 × 1440 pixel square (Supplementary Fig. 5).

Receptive field mapping

To map each neuron's receptive field, we presented local sinusoidal gratings (average 38 × 38°) drifting in 8 directions for 0.36 s each, across 48 overlapping locations, as previously⁹. Stimuli were full contrast, with an average spatial frequency of 0.14 cycles/° and a temporal frequency of 5 Hz. At each location, we calculated the local maximum spike frequency (red, inset, Fig. 1b, e). After subtracting the spontaneous rate, calculated for 0.8 s preceding stimulus onset (dotted, inset, Fig. 1b, e), we spatially interpolated the resulting local maximum responses 10 times (colour coding, Fig. 1a, d). Receptive field centres (red circle, Fig. 1b, e) and sizes (Supplementary Fig. 1f) were defined using the resulting 50% contour line (black, Fig. 1b, e). As recordings were performed with the animal ventral side up, receptive fields were rotated to display them with the dorsal side up (Supplementary Fig. 2a).

Next, at each location, we fit a cosine function to the responses to the different directions of motion, to extract the local preferred direction (LPD) and local motion sensitivity (LMS; inset, Fig. 1c, f). These were visualized as vectors where angle indicates LPD and length indicates LMS (arrows, Fig. 1a, d). We defined the receptive field preferred direction as the median of LPDs at locations where LMS exceeded 50% of the maximum (black and red arrows; Fig. 1c, f), using *circ_median* from the CircStat toolbox for MATLAB⁴¹. We calculated the LPD variance using the *circ_var* functions from CircStat toolbox for MATLAB⁴¹, of LPDs at locations where LMS exceeded 50% of the maximum (black and red arrows; Fig. 1c, f).

We extracted unpublished receptive field data from 100 reference neurons (other data from 5 of these optic flow sensitive descending neurons has been published previously⁴²) to set exclusion criteria (red shading, Supplementary Fig. 1a, b and c) and to classify the OFS DNs used in the rest of this study. OFS DN1 LHS neurons were defined by receptive field preferred directions between 120° and 200° (light green, Fig. 1g). OFS DN1 RHS neurons were defined by receptive field preferred directions between 340° and 60° (dark green, Fig. 1g). OFS DN2 neurons were classified by receptive field preferred directions between 220° and 320° (yellow and orange, Fig. 1g), with the location of the receptive field centre relative to the midline determining whether they were LHS or RHS (yellow and orange, Fig. 1h).

Directional sensitivity

Full-screen sinusoidal gratings were presented at full contrast, using the same spatial (0.14 cycles/°) and temporal (5 Hz) frequencies as in receptive field mapping. Directional responses were rotated to account for the ventral-side-up recording position, and data from LHS neurons were mirrored across the midline to display all neurons as RHS (Supplementary Fig 2a [↗](#)). For each stimulus direction, mean spike frequency was calculated over the stimulus duration, excluding the first 100 ms to avoid onset transients⁴³. A cosine function was fit to these responses, to extract its amplitude and preferred direction (Supplementary Fig 2b [↗](#)).

Responses to optic flow

We used a starfield stimulus to generate the type of perspective-corrected optic flow that would have been seen by the hoverfly if it was moving through a space of 2 cm diameter spheres (for details, see Ref.^{9,25}). These simulated translations at 0.5 m/s (sideslip, lift, and thrust, Supplementary movie 2) or rotations (pitch, yaw, and roll, Supplementary movie 1), at 50°/s. To quantify neural responses, the mean spike frequency was calculated over the 0.97 s stimulus duration, excluding the first 0.1 s to avoid onset-related transients⁴³. The spontaneous firing rate, averaged across 0.48 s immediately preceding stimulus onset (circles, Supplementary Fig 3a, b [↗](#)), was subtracted from the response.

Velocity response functions

We used four types of optic flow: three translations (sideslip, lift and thrust) and one rotation (roll). Translations were presented at velocities of -2, -1.5, -1, -0.4, -0.2, -0.1, 0, 0.1, 0.2, 0.4, 1, 1.5, and 2 m/s, while rotations were presented at 200, -150, -100, -40, -20, -10, 0, 10, 20, 40, 100, 150, and 200°/s. The sign of the velocity indicates the direction of motion as seen by the fly when corrected for its position, with positive values corresponding to counterclockwise roll, leftward sideslip, downward lift, and thrust moving away. Conversely, negative values indicate clockwise roll, rightward sideslip, upward lift, and thrust moving toward the hoverfly. The upper limit was defined by the movement of the individual dots within the starfield stimulus between frames, i.e. was limited by the refresh rate of the screen.

Each trial consisted of 39 stimuli (13 unique velocities × 3 repetitions), presented in a random order, with each stimulus lasting 2 s, immediately followed by the next velocity. Trials began with a 1 s blank screen, which served as the pre-stimulation baseline (Fig. 3a [↗](#) and Fig. 5a [↗](#)). Each optic flow condition was repeated multiple times, resulting in at least nine repetitions for each velocity and neuron, and five repetitions for each velocity and animal in behaviour.

For neuronal recordings, velocity response function trials were interleaved with the optic flow stimuli described above. In behavioural experiments, a flight refresher sequence interspersed each velocity tuning trial. This consisted of sinusoidal gratings (200° wavelength, 5 Hz) drifting rightward, leftward, and rightward again for 4 s each.

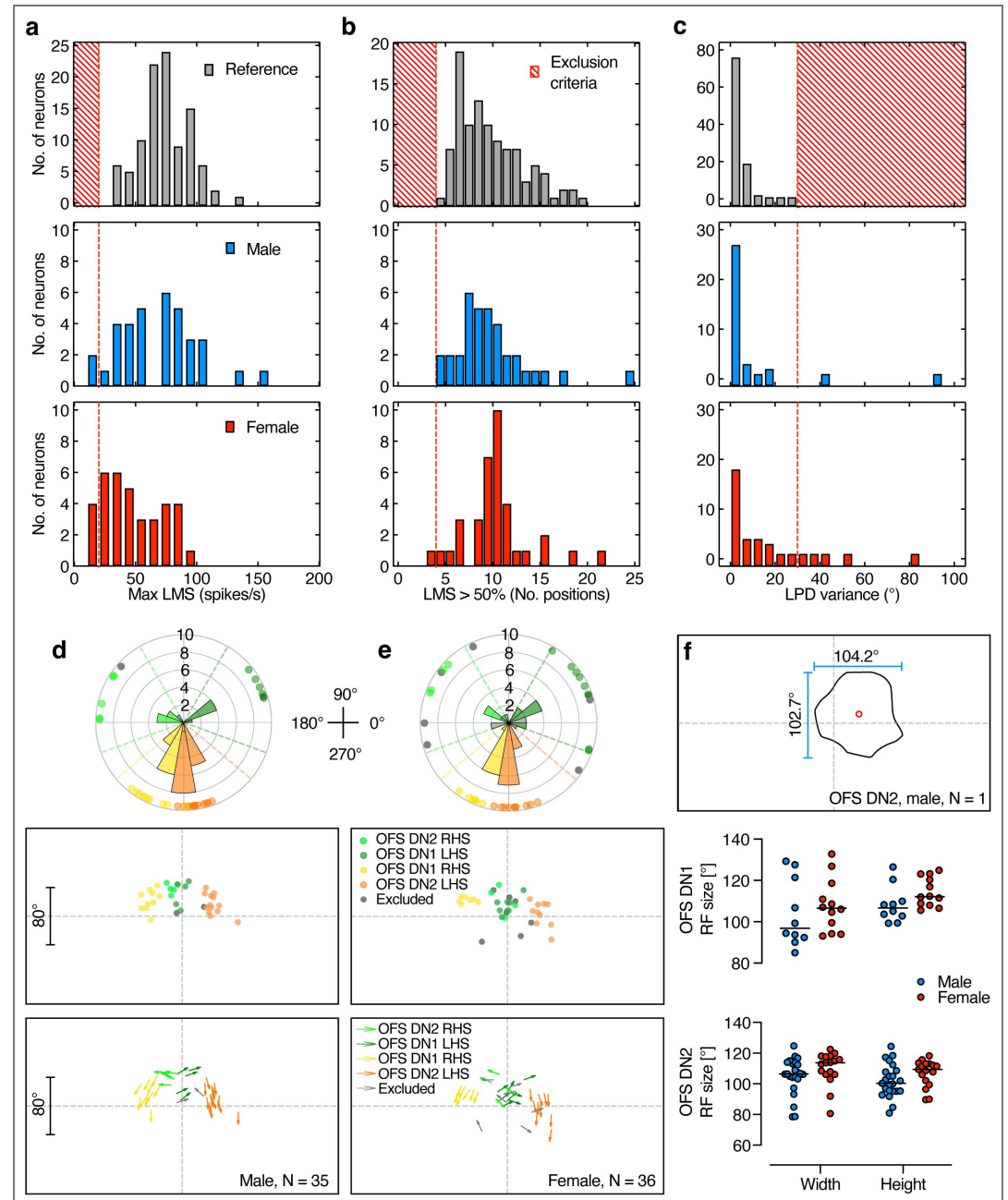
Neural responses were quantified as the mean spike frequency during the final second of stimulation (grey shaded areas, Fig. 3a–c [↗](#)). We then calculated the median across trials for each velocity (Fig. 3d [↗](#)). Pre-stimulation activity (spontaneous rate) was measured over a 2 s window immediately preceding stimulus onset. In behavioural experiments, we extracted the mean wing beat amplitude of the left and right wings ($\overline{WBA}_L$, $\overline{WBA}_R$, Fig. 5b [↗](#)) during the final second of stimulation (grey shaded areas, in Fig. 5 c, d [↗](#) and e [↗](#)). For each velocity, we calculated the WBA difference (WBAD, defined as $\overline{WBA}_L - \overline{WBA}_R$) and the WBA sum (WBAS, defined as $\overline{WBA}_L + \overline{WBA}_R$, Fig. 5b [↗](#)). We then calculated the median across repetitions for each individual (Fig. 5f [↗](#)). Pre-stimulation responses were averaged over a 0.5 s time window immediately preceding stimulus onset.

Response onset

For neuronal recordings, we compared onset times for clockwise roll ($+50^{\circ}/\text{s}$) and either upwards lift ($+0.5 \text{ m/s}$) for OFS DN 1 or downwards lift (-0.5 m/s) for OFS DN2, stimuli which generated strong responses in these neurons (Fig. 3 e, [f](#)). Mean spike frequency was calculated over the stimulus duration, excluding the first 0.1 s to avoid onset transients⁴³. Onset was defined as the first time point after the first 0.1 s, where spike rate exceeded 80% of the mean.

For behaviour, we used responses to roll ($-200^{\circ}/\text{s}$) and lift ($+2 \text{ m/s}$). Mean WBAS was defined as the average response during the final second of stimulation, and onset was defined as the first time point where WBAS exceeded 80% of this mean.

Supplementary information

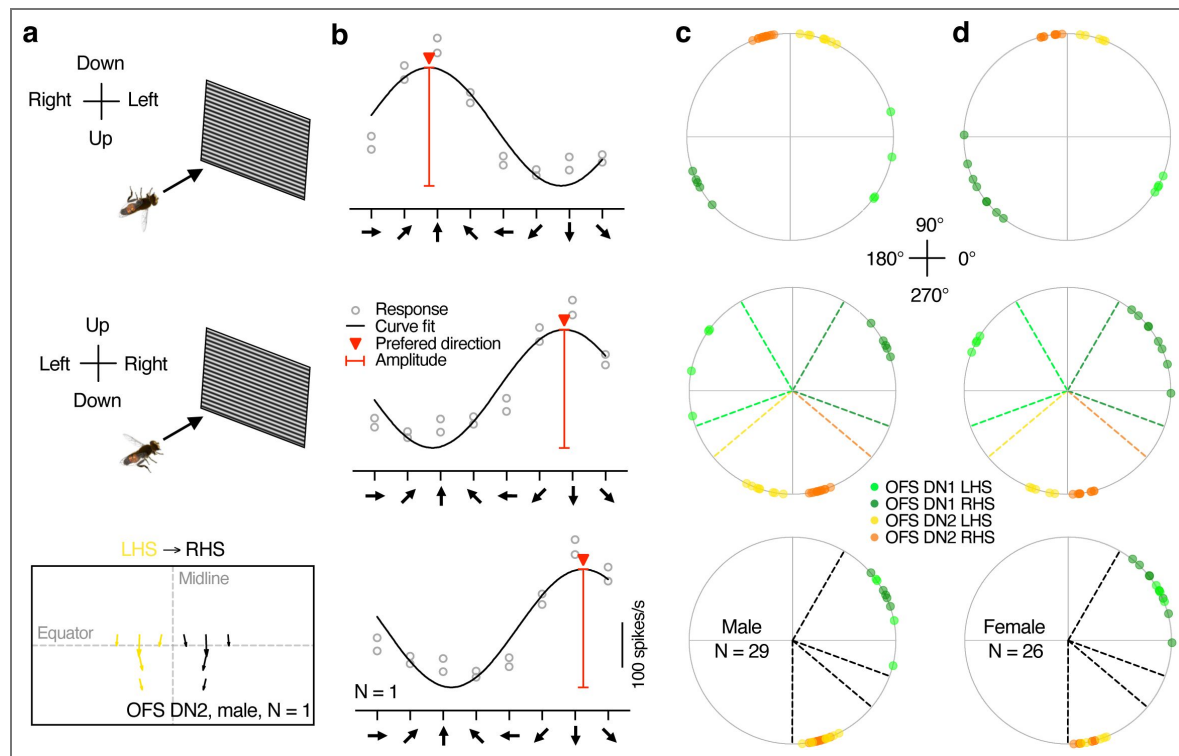


Supplementary Fig 1. Exclusion criteria and receptive field comparisons in male and female OFS DNs. a Distribution of maximum local motion sensitivity (LMS) in 100 male reference neurons (grey histogram, N = 100) compared to the neurons used for quantification in the rest of the paper (N = 35 males, 36 females). Six neurons with a maximum LMS below 20 spikes/s (red cross-hatched region or dashed line) were excluded from further analysis. **b** Number of stimulus positions with LMS greater than 50% of the neuron's maximum LMS (black and red arrows in Fig. 1c and f). Neurons with fewer than 4 positions were excluded. **c** Distribution of LPD variance. Neurons with directional variance exceeding 30° were excluded. **d** Classification of male neurons based on receptive field centre and preferred direction. The top panel shows the preferred directions across neurons, with dashed lines indicating the directional thresholds used for neuron type classification. The middle panel shows receptive field centre locations, and the bottom panel illustrates the relationship between receptive field centre and the receptive field preferred direction for all 35 male neurons, with 2 neurons excluded from further analysis.

(grey). **e** Same classification criteria applied to 36 female neurons, with 7 neurons excluded (grey). **f** The receptive field width and height at the 50% contour line for the remaining OFS DN 1 (N = 10 males, 12 females) and OFS DN2 (N = 23 males, 17 females). Individual data points shown, with black lines representing the median.

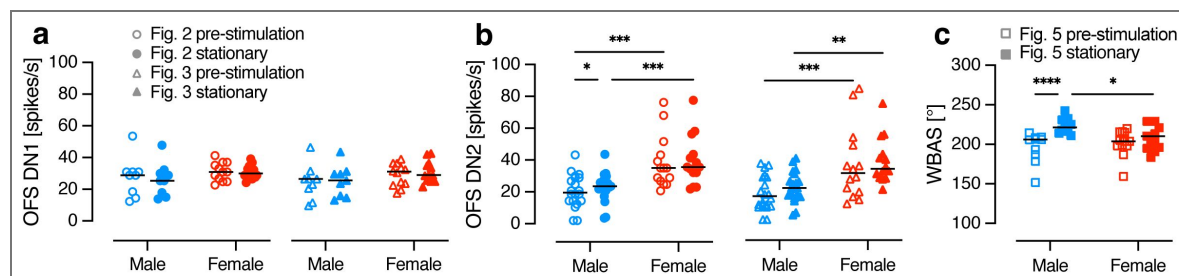
Supplementary Fig 2. Schematic illustration showing the steps used to standardize data across recordings.

a. Top panel, ventral-side-up position of hoverfly during electrophysiology recordings. Middle panel, reorientation to display the dorsal visual field at the top. Bottom panel, neurons with receptive fields in the left-hand side (LHS) of the visual field have been mirrored to the right-hand side (RHS). **b** Response of an example neuron to a full-screen, full-contrast sinusoidal grating (spatial wavelength 7°; temporal frequency 5 Hz) moving in eight different directions. The plots illustrate the effect of compensating for hoverfly orientation (middle panel) and receptive field location (bottom panel). The red arrowhead shows the neuron's preferred direction, and the red line indicates response amplitude for full-screen sinusoidal gratings. **c** Preferred direction of male hoverfly OFS DNs (N = 29). Neurons are colour coded by classification, with dashed lines indicating the receptive field thresholds. The plots show the effect of compensating the original data (top) for orientation (middle) and receptive field location (bottom). **d** Preferred direction of female hoverfly OFS DNs (N = 26), as in panel c.



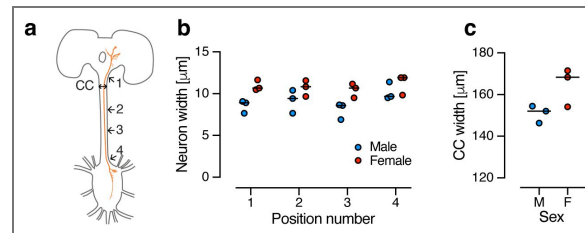
Supplementary Fig 3. Comparison of spontaneous activity and responses to stationary stimuli.

a Spontaneous activity (open shapes) and responses of OFS DN1 to the stationary starfield (filled shapes), from data extracting directional sensitivity (as in Fig. 2; circles) or velocity response functions (as in Fig. 3; triangles). **b** Spontaneous activity and responses of OFS DN2 as in panel a. **c** WBAS before stimulus presentation (open squares) or when viewing the stationary starfield (filled squares), quantified during velocity response function experiments (as in Fig. 5). Individual data points are shown, with lines representing the median. Asterisks denote statistical significance, two-way ANOVA with uncorrected Fisher's LSD test: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ and **** $p < 0.0001$.



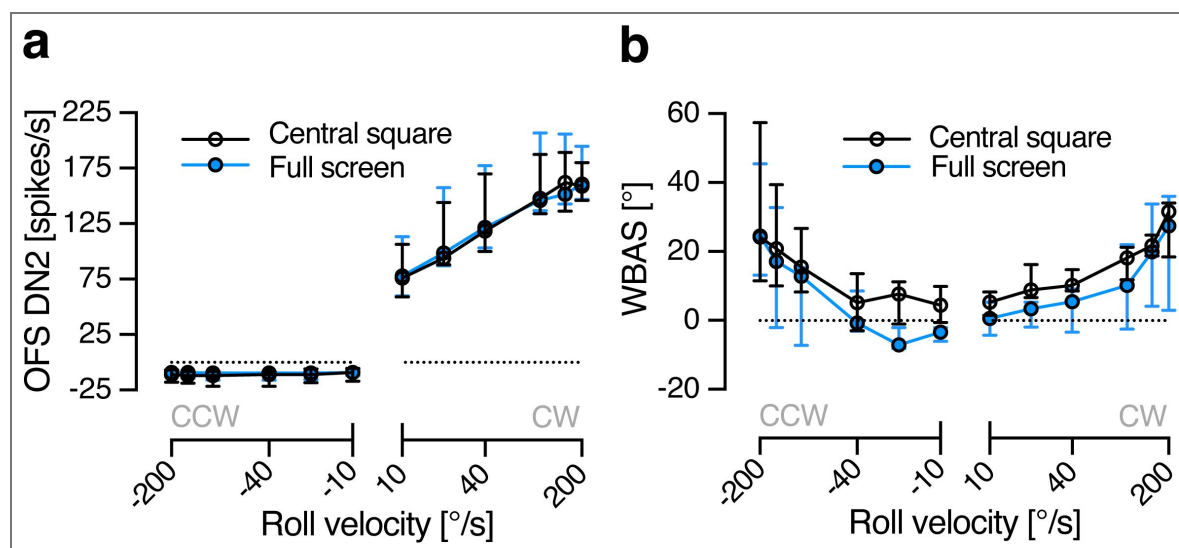
Supplementary Fig 4. Morphological quantification of OFS DN2.

a Schematic diagram of the hoverfly central brain and thoracic ganglia showing locations where the cervical connective (CC) and the OFS DN2 (orange) widths were measured. **b** Width measurements of OFS DN2 axons along the cervical connective at four anatomical landmarks: (1) immediately posterior to the central brain, (2) upper third of the CC, (3) lower third of the CC, and (4) immediately anterior to the thoracic ganglion. Data are shown for 3 male (blue) and 3 female (red) hoverflies. **c** Width of the anterior end of the cervical connective in the same individuals. All data are presented as individual repeats with the horizontal lines indicating median.



Supplementary Fig 5. Impact of stimulus size on neural and behavioural velocity response functions.

a Velocity response functions of male OFS DN2 neurons when the stimulus covered either the full screen (blue) or the central square (black, N = 7). **b** Velocity response functions of WBAS in male hoverflies when the stimulus covered either the full screen (blue) or the central square (black, N = 5). All data are presented as median and interquartile range.



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Additional information

Data availability and statistics

All data and analysis scripts have been submitted to DataDryad (10.5061/dryad.tb2rbp0fd). Throughout the text *n* refers to individual repetitions, whereas *N* refers to individual neurons (electrophysiology) or animals (behaviour). All data are presented as median and interquartile range, unless otherwise indicated. Statistical analyses were performed in Prism 10.4 (GraphPad Software), except for circular statistics, which were conducted using the CircStat toolbox for MATLAB⁴¹. The results of the statistical tests are given in figure legends and in Table 1 [↗](#).

Author contributions

Sarah Nicholas: conceptualization, methodology, validation, formal analysis, investigation, data curation, writing – original draft, visualization; **Katja Sporar Klinge:** methodology, validation, formal analysis, investigation, data curation, writing – review and editing, visualization; **Luke Turnbull:** methodology, validation, investigation, writing – review and editing; **Annabel Moran:** investigation, writing – review and editing; **Aika Young:** investigation, writing – review and editing; **Karin Nordström:** conceptualization, validation, formal analysis, resources, writing – original draft, visualization, supervision, project administration, funding acquisition; **Yuri Ogawa:** methodology, validation, formal analysis, investigation, data curation, writing – original draft, visualization, supervision, project administration.

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Additional files

Supplementary Movie 1. [🔗](#) Neural response to roll optic flow. The video shows the raw extracellular response of an OFS DN2 in red when the hoverfly was viewing an example roll stimulus, as in Fig. 3 of the main paper.

Supplementary Movie 2. [🔗](#) WBA response to sideslip optic flow. The video on the left shows the hoverfly filmed from above, with the coloured dots indicating the points tracked by DeepLabCut to extract the WBA. The stimulus the hoverfly was viewing is displayed on the right, as in Fig. 5 of the main paper.

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Peer reviews

Reviewer #1 (Public review):

Summary:

Hoverflies are known for a striking sexual dimorphism in eye morphology and early visual system physiology. Surprisingly, the male and female flight behaviors show only subtle differences. Nicholas et al. investigate the sensori-motor transformation of sexually dimorphic visual information to flight steering commands via descending neurons. The authors combined intra- and extracellular recordings, neuroanatomy, and behavioral analysis. They convincingly demonstrate that descending neurons show sexual dimorphisms - in particular at high optic flow velocities - while wing steering responses seem relatively monomorphic. The study highlights a very interesting discrepancy between neuronal and behavioral response properties.

More specifically, the authors focused on two types of descending neurons that receive inputs from well-characterized wide-field sensitive tangential cells: OFS DN1, which receives inputs from so-called HS cells, and OFS DN2, which receives input from a set of VS cells. Their likely counterparts in *Drosophila* connect to the neck, wing, and haltere neuropils. The authors characterized the visual response properties of these two neuronal classes in both male and female hoverflies and identified several interesting differences. They then presented the same set of stimuli, tracked wing beat amplitude, and analyzed the sum and the difference of right and left wing beat amplitude as a readout of lift or thrust, and yaw turning, respectively. Behavioral responses showed little to no sexual dimorphism, despite the observed neuronal differences.

Strengths:

I find the question very interesting and the results both convincing and intriguing. A fundamental goal in neuroscience is to link neuronal responses and behavior. The current study highlights that the transformations - even at the level of descending neurons to motoneurons - are complex and less straightforward than one might expect.

Weaknesses:

The authors investigated two types of descending neurons, but it was not clear to me how many other descending neurons are thought to be involved in wing steering responses to

wide-field motion. I would suggest providing a more in-depth overview of what is known about hoverflies and *Drosophila*, since the conclusions drawn from the study would be different if these two types were the only descending neurons involved, as opposed to representing a subset of the neurons conveying visual information to the wing neuropil.

Both neuronal classes have counterparts in *Drosophila* that also innervate neck motor regions. The authors filled the hoverfly DNs in intracellular recordings to characterize their arborization in the ventral nerve cord. In my opinion, these anatomical data could be further exploited and discussed a bit more: is the innervation in hoverflies also consistent with connecting to the neck and haltere motor regions? Are there any obvious differences and similarities to the *Drosophila* neurons mentioned by the authors? If the arborization also supports a role in neck movements, the authors could discuss whether they would expect any sexual dimorphism in head movements.

<https://doi.org/10.7554/eLife.109795.1.sa2>

Reviewer #2 (Public review):

Summary:

Many fly species exhibit male-specific visual behaviors during courtship, while little is known about the circuit underlying the dimorphic visuomotor transformations. Nicholas et al focus on two types of visual descending neurons (DNs) in hoverflies, a species in which only males exhibit high-speed pursuit of conspecifics. They combined electrophysiology and behavior analysis to identify these DNs and characterize their response to a variety of visual stimuli in both male and female flies. The results show that the neurons in both sexes have similar receptive fields but exhibit speed-dependent dimorphic responses to different optic flow stimuli.

Strengths:

Hoverflies, though not a common model system, show very interesting dimorphic behaviors and provide a unique and valuable entry point to explore the brain organization behind sexual dimorphism. The findings here are not only interesting on their own right but will also likely inspire those working in other systems, particularly *Drosophila*.

The authors employed rigorous morphology, electrophysiology, and behavior methods to deliver a comprehensive characterization of the neurons in question. The precision of the measurements allowed for identifying a subtle and nuanced neuronal dimorphism and set a standard for future work in this area.

Weaknesses:

Cell-typing using receptive field preferred directions (RFPDs): if I understood correctly, this classification method mostly relies on the LPDs near the center of the receptive field (median within the contour in Fig.1). I have two concerns here. First, this method is great if we are certain there are only two types of visual DNs as described in the manuscript. But how certain is this? Given the importance of vision in flight control, I would expect many DNs that transmit optic flow information to the motor center. I'd also like to point out that there are other lobula plate tangential cells (LPTCs) than HS and VS cells, which are much less studied and could potentially contribute to dimorphic behaviors. Second, this method feels somewhat impoverished given the richness of the data. The authors have nicely mapped out the directional tuning for almost the entire visual field. Instead of reducing this measurement to 2 values (center and direction), I was wondering if there is a better method to fully utilize the data at hand to get a better characterization of these DNs. As the authors are aware, local

features alone can be ambiguous in characterizing optic flows. What's more, taking into account more global features can be useful for discovering potentially new cell types.

Line 131, it wasn't clear to me why full-screen stimuli were used for comparison here, instead of the full receptive field maps. Male flies exhibit sexual dimorphic behaviors only during courtship, which would suggest that small-sized visual stimuli (mimicking an intruder or female conspecific) would be better suited to elicit dimorphic neuronal responses. A similar comment applies to the later results as well. Based on the receptive field mapping in Figure 1, I'm under the impression that these 2 DN types are more suited to detect wide-field optic flows, those induced by self-motion as mentioned in the manuscript. The results are still very interesting, but it's good to make this point clear early on to help set appropriate expectations. Conversely, this would also suggest that there are other visual DN types that are responsible for the courtship-related sexually dimorphic behaviors.

<https://doi.org/10.7554/eLife.109795.1.sa1>

Author response:

eLife Assessment

Hoverflies are known for their sexually dimorphic visual systems and exquisite flight behaviors. This valuable study reports how two types of visual descending neurons differ between males and females in their motion- and speed-dependent responses, yet surprisingly, the behavior they control lacks any sexual dimorphism. The results convincingly support these findings, which will be of interest for studies of visuomotor transformations and network-level brain organization.

This statement perfectly recapitulates our findings.

Public Reviews:

Reviewer #1 (Public review):

Summary:

Hoverflies are known for a striking sexual dimorphism in eye morphology and early visual system physiology. Surprisingly, the male and female flight behaviors show only subtle differences. Nicholas et al. investigate the sensori-motor transformation of sexually dimorphic visual information to flight steering commands via descending neurons. The authors combined intra- and extracellular recordings, neuroanatomy, and behavioral analysis. They convincingly demonstrate that descending neurons show sexual dimorphisms - in particular at high optic flow velocities - while wing steering responses seem relatively monomorphic. The study highlights a very interesting discrepancy between neuronal and behavioral response properties.

Thank you for this summary. Most of the statement perfectly recapitulates the main findings of our paper. However, we want to emphasize that some hoverfly flight behaviors are strongly sexually dimorphic, especially those related to courtship and mating. Indeed, only male hoverflies pursue targets at high speed, chase away territorial intruders, and pursue females for mating. However, other flight behaviours, such as those related to optomotor responses and flights between flowers when feeding, are not sexually dimorphic. We will amend the Introduction to make the difference between flight behaviors clear.

More specifically, the authors focused on two types of descending neurons that receive inputs from well-characterized wide-field sensitive tangential cells: OFS DN1, which receives inputs from so-called HS cells, and OFS DN2, which receives input from a set of VS cells. Their likely counterparts in Drosophila connect to the neck, wing, and haltere

neuropils. The authors characterized the visual response properties of these two neuronal classes in both male and female hoverflies and identified several interesting differences. They then presented the same set of stimuli, tracked wing beat amplitude, and analyzed the sum and the difference of right and left wing beat amplitude as a readout of lift or thrust, and yaw turning, respectively. Behavioral responses showed little to no sexual dimorphism, despite the observed neuronal differences.

Thank you for this very nice summary of our work. We want to clarify that LPTC input to DN1 and DN2 has not been shown directly in hoverflies using e.g. dye coupling, or dual recordings. Instead, the presumed HS and VS input is inferred from morphological and physiological DN evidence, and comparisons to similar data in *Drosophila* and blowflies. We will amend the Introduction to clarify this. The rest of the paragraph perfectly recapitulates the main findings of our paper.

Strengths:

I find the question very interesting and the results both convincing and intriguing. A fundamental goal in neuroscience is to link neuronal responses and behavior. The current study highlights that the transformations - even at the level of descending neurons to motoneurons - are complex and less straightforward than one might expect.

Thank you.

Weaknesses:

*The authors investigated two types of descending neurons, but it was not clear to me how many other descending neurons are thought to be involved in wing steering responses to wide-field motion. I would suggest providing a more in-depth overview of what is known about hoverflies and *Drosophila*, since the conclusions drawn from the study would be different if these two types were the only descending neurons involved, as opposed to representing a subset of the neurons conveying visual information to the wing neuropil.*

This is a great point. There are around 1000 fly DNs, of which many could respond to widefield motion, without being specifically tuned to widefield motion. For example, many looming sensitive neurons also respond to widefield motion, and could therefore be involved in the WBA movements that we measured here. In addition, there are many multimodal neurons that could be involved in optomotor responses in free flight, but these may not have been stimulated when we only provided visual input. Furthermore, many visual neurons are modulated by proprioceptive feedback, which is lacking in immobilized physiology preps. Finally, in blowflies, up to 5 optic flow sensitive DNs have been identified morphologically, and in *Drosophila* 3 have been identified morphologically and physiologically. In summary, it is more than likely that other neurons project visual widefield motion information to the wing neuropil. We will amend our Introduction and Discussion to make this important point clear to the readers.

*Both neuronal classes have counterparts in *Drosophila* that also innervate neck motor regions. The authors filled the hoverfly DNs in intracellular recordings to characterize their arborization in the ventral nerve cord. In my opinion, these anatomical data could be further exploited and discussed a bit more: is the innervation in hoverflies also consistent with connecting to the neck and haltere motor regions? Are there any obvious differences and similarities to the *Drosophila* neurons mentioned by the authors? If the arborization also supports a role in neck movements, the authors could discuss whether they would expect any sexual dimorphism in head movements.*

These are all great points. We did not see any clear arborizations to the frontal nerve, where we would expect to find the neck motor neurons (NMNs). In addition, while we did see fine

arborizations throughout the length of the thoracic ganglion, we saw no strong outputs projecting directly to the haltere nerve (HN). In the revised version of the MS we will modify figure 4 (morphological characterization) to clarify.

There are important differences between the morphology of DN1 and DN2 in hoverflies and DNHS1 and DNOVS2 in *Drosophila*, in terms of their projections in the thoracic ganglion. For example, In *Drosophila* DNOVS2, there are several fine branches along the length of the neuron in the thoracic ganglia. Similarly, we found fine branches in *Eristalis tenax* DN2, however, in addition, we found a wide branch projecting to the area of the thoracic ganglion where the prothoracic and pterothoracic nerves likely get their inputs (Figure 4), suggesting that the neuron could contribute to controlling the wings and/or the forelegs (which is why we quantified the WBA). In *Drosophila* DNHS1, there is a similar fat branch to the prothoracic and pterothoracic nerves, which we also found in *Eristalis tenax* OFS DN1 (Figure 4). Indeed, while *Drosophila* DNHS1 and DNOVS2 have quite strikingly different morphology, DN1 and DN2 in *Eristalis* looked quite similar. We will modify the Results section to make this clear.

In addition, to investigate this further, in the revised version of the MS we will include analysis of the movement of different body parts (including the head) to investigate the presence of any potential sexual dimorphism. Unfortunately, however, this will not include the halteres, as they cannot be seen well in the videos.

Reviewer #2 (Public review):

Summary:

Many fly species exhibit male-specific visual behaviors during courtship, while little is known about the circuit underlying the dimorphic visuomotor transformations. Nicholas et al focus on two types of visual descending neurons (DNs) in hoverflies, a species in which only males exhibit high-speed pursuit of conspecifics. They combined electrophysiology and behavior analysis to identify these DNs and characterize their response to a variety of visual stimuli in both male and female flies. The results show that the neurons in both sexes have similar receptive fields but exhibit speed-dependent dimorphic responses to different optic flow stimuli.

This statement perfectly recapitulates the main findings of our paper. However, as mentioned above, while hoverfly flight behaviors related to courtship and mating are strongly sexually dimorphic, other flight behaviours, such as those related to optomotor responses and flights between flowers when feeding, are not. We will amend the Introduction to make the difference between flight behaviors clear.

Strengths:

*Hoverflies, though not a common model system, show very interesting dimorphic behaviors and provide a unique and valuable entry point to explore the brain organization behind sexual dimorphism. The findings here are not only interesting on their own right but will also likely inspire those working in other systems, particularly *Drosophila*.*

Thank you.

The authors employed rigorous morphology, electrophysiology, and behavior methods to deliver a comprehensive characterization of the neurons in question. The precision of the measurements allowed for identifying a subtle and nuanced neuronal dimorphism and set a standard for future work in this area.

Thank you.

Weaknesses:

Cell-typing using receptive field preferred directions (RFPDs): if I understood correctly, this classification method mostly relies on the LPDs near the center of the receptive field (median within the contour in Fig. 1). I have two concerns here. First, this method is great if we are certain there are only two types of visual DNs as described in the manuscript. But how certain is this? Given the importance of vision in flight control, I would expect many DNs that transmit optic flow information to the motor center. I'd also like to point out that there are other lobula plate tangential cells (LPTCs) than HS and VS cells, which are much less studied and could potentially contribute to dimorphic behaviors.

This is very true, and an important point. As mentioned above, in blowflies, up to 5 optic flow sensitive DNs have been identified morphologically, however, if these correspond to 5 different physiological types remain unclear. In both blowflies and *Drosophila* 3 have been identified morphologically and physiologically (DNHS1, DNOVS1, DNOVS2). Importantly, in both blowflies and fruitflies DNOVS1 gives graded responses, and no action potentials, meaning that we would not be able to record from it using extracellular electrophysiology.

We previously used clustering techniques to show that in *Eristalis*, we can reliably distinguish two types of optic flow sensitive DNs from extracellular electrophysiological data, based on a range of receptive field parameters, and we think that these correspond to DNHS1 and DNOVS2 in *Drosophila* (Nicholas et al, J Comp Physiol A, 2020, cited in paper). As mentioned above in response to Reviewer 1, this does not mean that there are no other neurons that could respond to widefield optic flow, and which might be involved in the WBA we recorded in the paper. However, the point of this paper was not to conclusively show that there are only two optic flow sensitive descending neurons. The point was to say that there are two quite distinct optic flow sensitive neurons that have similar receptive fields in males and females, while the responses to widefield motion show differences between males and females.

We will modify the Introduction and Discussion to make these important points clear to the Reader, including the discussion of the 45-60 LPTCs that exist in the lobula plate, and what their role might be.

Second, this method feels somewhat impoverished given the richness of the data. The authors have nicely mapped out the directional tuning for almost the entire visual field. Instead of reducing this measurement to 2 values (center and direction), I was wondering if there is a better method to fully utilize the data at hand to get a better characterization of these DNs. As the authors are aware, local features alone can be ambiguous in characterizing optic flows. What's more, taking into account more global features can be useful for discovering potentially new cell types.

This is a great point, and we did an extensive analysis of other receptive field properties in this study (shown in supp fig 1). In addition, and as mentioned above, we have published a clustering analysis across receptive field properties of these neurons (Nicholas et al, J Comp Physiol A, 2020, cited in paper). The point that we attempted to make in this paper was that by using two strikingly simple metrics, we can reliably distinguish which of the two neuron types we are recording from (if we accept that there are two main types that we are likely to record from) simply based on location and overall directional preference. This makes automated analysis very easy and straightforward. Indeed, we now use this routinely to ID what neuron we are recording from, rather than making a human-based assumption.

However, we agree that further in depth analysis is warranted. Therefore, to address this, we will provide additional receptive field analysis and clustering in the revised version of the MS. In addition, we want to highlight that all data is uploaded to DataDryad for anyone interested in doing additional in-depth analyses.

Line 131, it wasn't clear to me why full-screen stimuli were used for comparison here, instead of the full receptive field maps. Male flies exhibit sexual dimorphic behaviors only during courtship, which would suggest that small-sized visual stimuli (mimicking an intruder or female conspecific) would be better suited to elicit dimorphic neuronal responses. A similar comment applies to the later results as well. Based on the receptive field mapping in Figure 1, I'm under the impression that these 2 DN types are more suited to detect wide-field optic flows, those induced by self-motion as mentioned in the manuscript. The results are still very interesting, but it's good to make this point clear early on to help set appropriate expectations. Conversely, this would also suggest that there are other visual DN types that are responsible for the courtship-related sexually dimorphic behaviors.

Thank you for mentioning these important points. Our reasoning for using full-screen stimuli for the analysis on line 131 was that since we used the small sinusoidal gratings for mapping the receptive fields, and to subsequently classify the neurons, it would be unfair to use the same data to investigate potential sexual dimorphism. I.e., we selected neurons that fulfilled certain criteria, and then we cannot rightfully use the same criteria to determine differences. This was not explicitly mentioned in the paper, so we will modify the text to make this clear to the Reader.

However, in Supp Figure 1d/e we show that there are no striking receptive field differences between males and females in terms of receptive field center nor directional preference. In Supp Figure 1f we show that there is no difference between male and female receptive field height and width. We will modify the text to draw the Reader's attention to this figure, and also mention the additional analysis done in response to the comment above.

As a side note, I personally expected at least DNHS1 to have a smaller receptive field in males, as the hoverfly HSN is strikingly sexually dimorphic (Nordström et al, Curr Biol 2008), and also very sensitive to small objects. However, while optic flow sensitive DNs do respond to small objects (see e.g. the J Comp Physiol paper mentioned above) we did not detect any obvious sexual dimorphism in receptive field properties. Indeed, we think that a different subset of DNs control target pursuit behavior (target selective DNs (TSDNs)). This will be addressed in the modified version of the paper.

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